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Course Title: Software Configuration Mangement Technology

Course Code: ST-1206

Class: BS SET 2ND Semester (2025-2029)

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Department: Software Engineering

Lab Tasks

Lab Task 1 # Install Git & Configure Identify Setting

Git Version:

```
PS C:\Users\PMLS> git --version
git version 2.53.0.windows.1
PS C:\Users\PMLS>
```

Username Configuration:

```
PS C:\Users\PMLS> git config --global user.name "Ameen-UL-Husnaat"
PS C:\Users\PMLS> git
```

User Email Configuration:

```
PS C:\Users\PMLS> git config --global user.email "ameenulhusnat82119@gmail.com"
PS C:\Users\PMLS>
```

Verify Setting:

```
PS C:\Users\PMLS> git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
filter.lfs.clean=git-lfs clean -- %f
user.name=Ameen-UL-Husnaat
user.email=ameenulhusnat82119@gmail.com
gui.recentrepo=D:/husnaat/University work/2nd semester/Software Configuration Management Technology/Lab_Task_02
gui.recentrepo=C:/Users/PMLS/scmt-week2-Ameen UL Husnaat
core.editor=code --wait
PS C:\Users\PMLS>
```

Lab Task 2 # First Repository & Git Workflow

Git Initializing:

```
PS C:\Users\PMLS> cd SCMT_Lab2_2025_SET_05
PS C:\Users\PMLS\SCMT_Lab2_2025_SET_05> git init
Initialized empty Git repository in C:/Users/PMLS/SCMT_Lab2_2025_SET_05/.git/
PS C:\Users\PMLS\SCMT_Lab2_2025_SET_05>
```

Git Check Status:

```
PS C:\Users\PMLS\SCMT_Lab2_2025_SET_05> git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
PS C:\Users\PMLS\SCMT_Lab2_2025_SET_05>
```

Git Status Untracked, Staged, Clean:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git add README.md
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md
```

Git Commit Output:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git log --oneline
2d4e9c5 (HEAD -> master) print Good by message
9eccc54 Initial commit: add main.py file and print Hello World message
4b6803c Initial commit: add README file
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05>
```

Git Commit History:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git commit -am "print Good by message"
[master 2d4e9c5] print Good by message
1 file changed, 2 insertions(+), 1 deletion(-)
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git log
commit 2d4e9c52854d453ecf7f9821343da265e3f20186 (HEAD -> master)
Author: Ameen-UL-Husnaat <ameenulhusnat82119@gmail.com>
Date: Fri May 15 19:01:13 2026 +0500

    print Good by message

commit aeccf54645cdd16de7cb488225f9fa8753d8928e
Author: Ameen-UL-Husnaat <ameenulhusnat82119@gmail.com>
Date: Fri May 15 18:51:17 2026 +0500

    Initial commit: add main.py file and print Hello World message

commit 4b6803c2fa440ff9d8b4d5774b701c874ef37f2f
Author: Ameen-UL-Husnaat <ameenulhusnat82119@gmail.com>
Date: Fri May 15 18:30:30 2026 +0500

    Initial commit: add README file
```

Git Commit Details:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git log --stat
commit 2d4e9c52854d453ecf7f9821343da265e3f20186 (HEAD -> master)
Author: Ameen-UL-Husnaat <ameenulhusnat82119@gmail.com>
Date: Fri May 15 19:01:13 2026 +0500

    print Good by message

    main.py | 3 ++-
    1 file changed, 2 insertions(+), 1 deletion(-)

commit aeccf54645cdd16de7cb488225f9fa8753d8928e
Author: Ameen-UL-Husnaat <ameenulhusnat82119@gmail.com>
Date: Fri May 15 18:51:17 2026 +0500

    Initial commit: add main.py file and print Hello World message

    main.py | 1 +
    1 file changed, 1 insertion(+)

commit 4b6803c2fa440ff9d8b4d5774b701c874ef37f2f
Author: Ameen-UL-Husnaat <ameenulhusnat82119@gmail.com>
Date: Fri May 15 18:30:30 2026 +0500

    Initial commit: add README file
```

Lab Task 3 # Branching & Merging

Git Branch Creation:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git branch
* master
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git checkout -b feature/profile
Switched to a new branch 'feature/profile'
```

Git Current Branch:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git branch
* feature/profile
  master
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> |
```

Git Commit on Feature Branch:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git add profile.txt
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git commit -m "Initial commit: add profile.txt file"
[feature/profile c28d1b7] Initial commit: add profile.txt file
1 file changed, 2 insertions(+)
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git add profile.txt
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git commit -m "Update profile with name and roll no"
[feature/profile 97f737a] Update profile with name and roll no
1 file changed, 1 insertion(+)
```

Git Switch Main Branch:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git add profile.txt
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git commit -m "Update profile with name and roll no"
1 file changed, 1 insertion(+)
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git checkout main
error: pathspec 'main' did not match any file(s) known to git
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git checkout master
Switched to branch 'master'
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git branch -m master main
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git checkout main
Already on 'main'
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> |
```

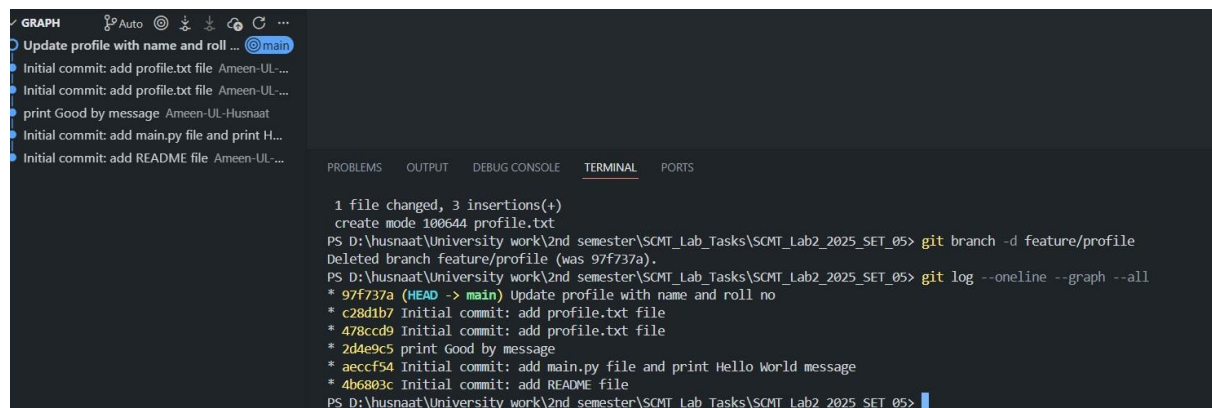

Git Merge Output:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git merge feature/profile
Updating 2d4e9c5..97f737a
Fast-forward
 profile.txt | 3 +++
1 file changed, 3 insertions(+)
create mode 100644 profile.txt
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05>
```

Git Branch Deletion:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git branch -d feature/profile
Deleted branch feature/profile (was 97f737a).
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05>
```

Git Merge History:



The screenshot shows the VS Code interface with the Git merge history graph on the left and the terminal output on the right. The graph shows a sequence of commits: 'Update profile with name and roll ...' (main), 'Initial commit: add profile.txt file' (Ameen-UL-...), 'Initial commit: add profile.txt file' (Ameen-UL-...), 'print Good by message' (Ameen-UL-Husnaat), 'Initial commit: add main.py file and print H...', and 'Initial commit: add README file' (Ameen-UL-...). The terminal output shows the following commands and results:

```
1 file changed, 3 insertions(+)
create mode 100644 profile.txt
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git branch -d feature/profile
Deleted branch feature/profile (was 97f737a).
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git log --oneline --graph --all
* 97f737a (HEAD -> main) Update profile with name and roll no
* c28d1b7 Initial commit: add profile.txt file
* 478ccd9 Initial commit: add profile.txt file
* 2d4e9c5 print Good by message
* aeccf54 Initial commit: add main.py file and print Hello World message
* 4b6803c Initial commit: add README file
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05>
```

Lab Task 4 # Git Stash & Undoing Changes

Git Stash:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git stash
Saved working directory and index state WIP on main: 97f737a Update profile with name and roll no
```

Git Stash List:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git stash list
stash@{0}: WIP on main: 97f737a Update profile with name and roll no
```

Git Stash Pop:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git stash pop
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   profile.txt

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (6662eee8bb6840ecdabd2fd8781ea76514f8ae75)
```

Git Reset –Soft & Status Afterward:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git reset --soft HEAD~1
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   profile.txt
```

Git Revert Result & Commit Amend:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git revert 72c0ae3
hint: Waiting for your editor to close the file...
[10788:0515/214011.524:ERROR:components\services\storage\service_worker\service_worker_storage.cc:1809] Failed to delete the database:
Database IO error
[main 10192d7] gitRevert "modified profile.txt"
1 file changed, 1 deletion(-)
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git revert 72c0ae3 --no-edit
On branch main
nothing to commit, working tree clean
```

Git Reflog Output:

```
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> git log --oneline
10192d7 (HEAD -> main) gitRevert "modified profile.txt"
72c0ae3 modified profile.txt
97f737a Update profile with name and roll no
c28d1b7 Initial commit: add profile.txt file
478ccd9 Initial commit: add profile.txt file
2d4e9c5 print Good by message
aeccf54 Initial commit: add main.py file and print Hello World message
4b6803c Initial commit: add README file
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\SCMT_Lab2_2025_SET_05> |
```

THE END
